

Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector quantity**

Forces have both magnitude (strength) and direction.

- **SI unit**

The standard metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of 1 m/s².

- **Formula**

Newton's Second Law of Motion:

$$F = m \cdot a$$

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where

(F) = force,

(m) = mass,

(a) = acceleration.

Common Types of Forces

Type	Description	Examples
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• Contact forces	Applied through physical contact between objects	Friction, tension, air resistance, a push or pull
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• Non-contact forces	Act over a distance without physical contact	Gravity, electromagnetism, magnetic forces, nuclear forces
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> • Note:

> - Forces are always represented as vectors, so direction matters.

> - The SI unit “Newton” is derived from the fundamental units: $1\text{ N} = 1\text{ kg}\cdot\text{m/s}^2$.

> - Newton’s Second Law links force, mass, and acceleration, providing a quantitative way to analyze motion.